

Local deployment tutorial

An online demonstration of the platform is available at <http://47.115.46.121:5000/>. However, due to limitations in data persistence and computational capacity, local deployment is strongly recommended for sustained use and full functionality.

1. Development Environment Setup

To begin, ensure that the necessary development environments are properly configured, specifically for Python and C/C++.

For Python, it is highly recommended to use Anaconda for streamlined management. Download and install Anaconda from the official website (<https://www.anaconda.com/download>). Upon installation, the Python runtime environment will be readily available.

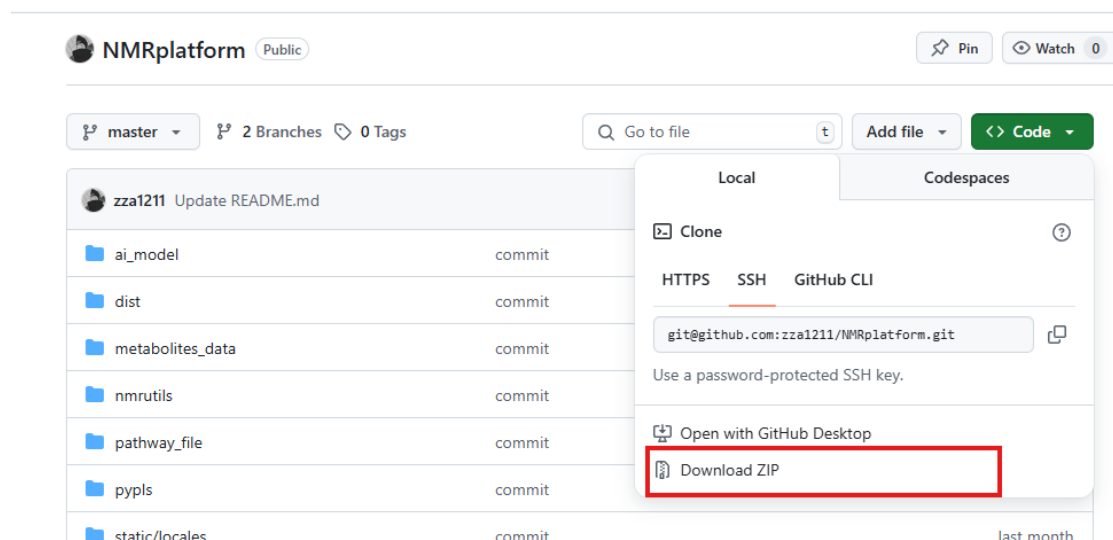
For the C/C++ environment, the setup varies by operating system. On Windows, install Visual Studio from <https://visualstudio.microsoft.com/>. For Linux systems, you can simply execute the following command in the terminal:

```
bash
```

```
sudo apt install gcc
```

2. Code Deployment

Begin by downloading the platform's source code from GitHub as a ZIP archive.



After extracting the files, open a terminal and install the required Python dependencies using:

```
Pip install -r requirements.txt
```

Next, set up the runtime environment for the deep learning models by executing:

```
conda install pytorch==1.11.0 torchvision==0.12.0 torchaudio==0.11.0 cpuonly -c pytorch
```

The platform utilizes the pypls library for PLS-DA and OPLS-DA analyses. To install it, follow the instructions at <https://github.com/Omicometrics/pypls/tree/master>. Specifically:

- I. Download and extract the pypls source code.
- II. Navigate to the extracted directory.
- III. Run the following command to complete the installation:

```
python -m setup install
```

3. OpenAI API Key Configuration

To access large language model services, an OpenAI API key is required.

- I. Register and log in at <https://platform.openai.com/docs/overview>.
- II. Generate a new API key at <https://platform.openai.com/api-keys>.
- III. Open the app.py file and locate Line 110.
- IV. Replace the placeholder with your newly generated API key.

Finally, launch the platform by executing:

```
python app.py
```

4. Troubleshooting Deployment Issues

- I. **FlaskUpload Import Error:** If you encounter the error `cannot import name 'secure_filename' from werkzeug.utils`, modify the import statement:

```
from werkzeug.utils import secure_filename
```

- II. **FileStorage Import Error:** If the error `cannot import name 'FileStorage' from werkzeug.datastructures` arises, separate the import into two lines:

```
from werkzeug.utils import secure_filename
```

```
from werkzeug.datastructures import FileStorage
```